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PART 1238—MICROFORMS RECORDS MANAGEMENT

Authority: [44 U.S.C. chapters 29 and 33](#).

Source: [74 FR 51014](#), Oct. 2, 2009, unless otherwise noted.

Subpart A—General

§ 1238.1 What is the scope of this part?

This part covers the standards and procedures for using micrographic technology in the management of Federal records.

§ 1238.2 What are the authorities for part 1238?

The statutory authorities for this part are [44 U.S.C. chapters 29 and 33](#).

§ 1238.3 What definitions apply to this part?

See [§ 1220.18 of this subchapter](#) for definitions of terms used in part 1238.

§ 1238.4 What standards are used as guidance for this part?

These regulations conform with guidance provided in ISO15489-1:2001, part 7.1 (Principles of records management programmes), and 9.6 (storage and handling).

§ 1238.5 What publications are incorporated by reference in this part?

- (a) NARA incorporates certain material by reference into this part with the approval of the Director of the Federal Register under [5 U.S.C. 552\(a\)](#) and [1 CFR part 51](#). To enforce any edition other than that specified in this section, NARA must publish a document in the FEDERAL REGISTER. The material must be available to the public. You may inspect all

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approved material incorporated by reference at NARA's textual research room, located at National Archives and Records Administration; 8601 Adelphi Road; Room 2000; College Park, MD 20740-6001. To arrange to inspect this approved material at NARA, contact NARA's Regulation Comments Desk (Strategy and Performance Division (MP)) by email at regulation_comments@nara.gov or by telephone at 301.837.3151. All approved material is also available from the sources listed below. You may also inspect approved material at the Office of the Federal Register (OFR). For information on the availability of this material at the OFR, call 202.741.6000 or go to http://www.archives.gov/federal_register/cfr/ibr-locations.html.

- (b) **American National Standards Institute (ANSI) and International Organization for Standards (ISO) standards.** The following ANSI and ISO standards are available from the American National Standards Institute, 25 West 43rd St., 4th Floor, New York, NY 10036, phone number (212) 642-4900, or online at <http://webstore.ansi.org>.

- (1) ANSI/AIIM MS1-1996 ("ANSI/AIIM MS1"), Standard Recommended Practice for Alphanumeric Computer-Output Microforms—Operational Practices for Inspection and Quality Control, August 8, 1996, IBR approved for § 1238.14.
- (2) ANSI/AIIM MS5-R1998 ("ANSI/AIIM MS5"), Standard for Information and Image Management—Microfiche, December, 1998, IBR approved for § 1238.10.
- (3) ANSI/AIIM MS14-1996 ("ANSI/AIIM MS14"), Standard Recommended Practice—Specifications for 16mm and 35mm Roll Microfilm, August 8, 1996, IBR approved for § 1238.10.
- (4) ANSI/AIIM MS19-1993 ("ANSI/AIIM MS19"), Standard Recommended Practice—Identification of Microforms, August 18, 1993, IBR approved for § 1238.12.
- (5) ANSI/AIIM MS32-1996 ("ANSI/AIIM MS32"), Standard Recommended Practice—Microrecording of Engineering Source Documents on 35mm Microfilm, February 16, 1996, IBR approved for § 1238.10.
- (6) ANSI/AIIM MS41-1996 ("ANSI/AIIM MS41"), Dimensions of Unitized Microfilm Carriers and Apertures (Aperture, Camera, Copy and Image Cards), July 16, 1996, IBR approved for § 1238.10.
- (7) ANSI/AIIM MS43-1998 ("ANSI/AIIM MS43"), Standard Recommended Practice—Operational Procedures—Inspection and Quality Control of Duplicate Microforms of Documents and From COM), June 2, 1998, IBR approved for § 1238.14,
- (8) ANSI/AIIM MS45-1990 ("ANSI/AIIM MS 45"), Recommended Practice for Inspection of Stored Silver-Gelatin Microforms for Evidence of Deterioration, January 22, 1990, IBR approved for § 1238.22.
- (9) ISO 18911:2000 ("ISO 18911"), Imaging materials—Processed safety photographic films—Storage practices, First Edition, November 1, 2000, IBR approved for § 1238.20.

- (c) **Techstreet.** The following standards are available from the standards reseller Techstreet, 3916 Ranchero Drive, Ann Arbor, MI 48108, phone number (800) 699-9277, or online at www.Techstreet.com.

- (1) ISO 18901:2002 ("ISO 18901"), Imaging Materials—Processed silver-gelatin type black-and-white films—Specifications for stability, February 15, 2002, IBR approved for §§ 1238.10, 1238.14, and 1238.20.

- (2) Reserved

- (d) **Document Center Inc.** The following are available from the standard reseller the Document Center Inc., 111 Industrial Road, Suite 9,

- (1) ANSI/NAPM IT2.19-1994 ("ANSI/NAPM IT2.19), American National Standard for Photography—Density Measurements—Part 2: Geometric Conditions for Transmission Density, February 20, 1995, IBR approved for § 1238.14.
- (2) ANSI/PIMA IT9.2-1998 ("ANSI/PIMA IT9.2"), Photographic Processed Films, Plates, and Papers—Filing Enclosures and Storage Containers, April 15, 1998, IBR approved for §§ 1238.10 and 1238.20.
- (3) ANSI/AIIM MS 23-1998 ("ANSI/AIIM MS23"), Standard Recommended Practice—Production, Inspection, and Quality Assurance of First-Generation, Silver Microforms of Documents, June 2, 1998, IBR approved for §§ 1238.10 and 1238.14.
- (4) ANSI/ISO 3334-1991, ANSI/AIIM MS51-1991 ("ANSI/ISO 3334, ANSI/AIIM MS51"), Micrographics—ISO Resolution Test Chart No. 2—Description and Use, May 10, 1991, IBR approved for § 1238.14.
- (5) ANSI/NAPM IT2.18-1996 ("ANSI/NAPM IT2.18"), American National Standard for Photography—Density Measurements—Part 3: Spectral Conditions, March 8, 1996, IBR approved for § 1238.14.

[74 FR 51014, Oct. 2, 2009, as amended at 83 FR 13657, Mar. 30, 2018]

◉ Subpart B—Microfilming Standards

◉ § 1238.10 What are the format standards for microfilming records?

The following formats must be used when microfilming records:

(a) **Roll film** —

- (1) **Source documents.** The formats described in ANSI/AIIM MS14 (incorporated by reference, see § 1238.5) must be used for microfilming source documents on 16mm and 35mm roll film. A reduction ratio no greater than 1:24 is recommended for correspondence or similar typewritten documents. Use ANSI/AIIM MS23 (incorporated by reference, see § 1238.5) for the appropriate reduction ratio and format for meeting image quality requirements. When microfilming on 35mm film for aperture card applications, the format dimensions in ANSI/AIIM MS32 (incorporated by reference, see § 1238.5), Table 1 must be used, and the aperture card format "D Aperture" shown in ANSI/AIIM MS41 (incorporated by reference, see § 1238.5), Figure 1, must be used. The components of the aperture card, including the paper and adhesive, must conform to the requirements of ANSI/PIMA IT9.2 (incorporated by reference, see § 1238.5). The 35mm film used in the aperture card application must conform to film designated as LE 500 in ISO 18901 (incorporated by reference, see § 1238.5).
- (2) **COM.** Microfilm created using computer output microfilm (COM) technology must use the simplex mode described in ANSI/AIIM MS14 (incorporated by reference, see § 1238.5) at an effective ratio of 1:24 or 1:48 depending upon the application.

- (b) **Microfiche.** When creating microfiche, either by microfilming source documents or using COM technology, the formats and reduction ratios prescribed in ANSI/AIIM MS5 (incorporated by reference, see § 1238.5) must be used as specified for the size and quality of the documents being filmed. Use ANSI/AIIM MS23 (incorporated by reference, see § 1238.5) for determining the appropriate reduction ratio and format for meeting the image quality requirements.

(c) **Index placement** —

- (1) **Source documents.** When microfilming source documents, place indexes, registers, or other finding aids, if microfilmed, either in the first frames of the first roll of film or in the last frames of the last roll of film of a series. For microfiche, place the indexes in the last frames of the last microfiche or microfilm jacket of a series.
- (2) **COM.** Place indexes on COM following the data on a roll of film, in the last frames of a single microfiche, or in the last frames of the last fiche in a series. Other locations for indexes may be used only if dictated by special system constraints.

◉ § 1238.12 What documentation is required for microfilmed records?

Agencies must ensure that the microforms capture all information contained on the source documents and that they can be used for the purposes the source documents served. Microform records must be labeled and organized to support easy retrieval and use. Agencies must:

- (a) Arrange, describe, and index the filmed records to permit retrieval of any particular document or component of the records.
- (b) Title each microform roll or fiche with a titling target or header. For fiche, place the titling information in the first frame if the information will not fit on the header. At a minimum, titling information must include:
 - (1) The title of the records;
 - (2) The number or identifier for each unit of microform;
 - (3) The security classification, if any; and
 - (4) The name of the agency and sub-organization, the inclusive dates, names, or other data identifying the records to be included on a unit of microform.
- (c) Add an identification target showing the date of microfilming. When necessary to give the microform copy legal standing, the target must also identify the person who authorized the microfilming. Use ANSI/AIIM MS19 (incorporated by reference, see § 1238.5) for standards for identification targets.

◉ § 1238.14 What are the microfilming requirements for permanent and unscheduled records?

- (a) Agencies must apply the standards in this section when microfilming:
 - (1) Permanent paper records where the original paper record will be destroyed (only after authorization from NARA);
 - (2) Unscheduled paper records where the original paper record will be destroyed (only after authorization from NARA); and
 - (3) Permanent and unscheduled original microform records (no paper originals) produced by automation, such as COM.
- (b) Agencies must use polyester-based silver gelatin type film that conforms to ISO 18901 (incorporated by reference, see § 1238.5) for LE 500 film in all applications.
- (c) Agencies must process microforms so that the residual thiosulfate ion concentration will not exceed 0.014 grams per square meter in accordance with ISO 18901 (incorporated by reference, see § 1238.5) and use the processing procedures in ANSI/AIIM MS1 and ANSI/AIIM MS23 (both incorporated by reference, see § 1238.5).
- (d) Agencies must use the following standards for quality:
 - (1) **Resolution** —
 - (i) **Source documents.** Agencies must determine minimum resolution on microforms of source documents using the

method in the Quality Index Method for determining resolution and anticipated losses when duplicating, as described in ANSI/AIIM MS23 and ANSI/AIIM MS43 (both incorporated by reference, see § 1238.5). Agencies must perform resolution tests using an ANSI/ISO 3334 Resolution Test Chart (incorporated by reference, see § 1238.5) or a commercially available certifiable target manufactured to comply with this standard, and read the patterns following the instructions of ANSI/ISO 3334. Agencies must use the smallest character used to display information to determine the height used in the Quality Index formula. Agencies must use a Quality Index of five at the third generation level.

- (ii) **COM.** COM must meet the requirements of ANSI/AIIM MS1 (incorporated by reference, see § 1238.5).

(2) **Background density of images.** Agencies must use the background ISO standard visual diffuse transmission density on microforms appropriate to the type of documents being filmed. Agencies must use the procedure for density measurement described in ANSI/AIIM MS23 (incorporated by reference, see § 1238.5). The densitometer must meet with ANSI/NAPM IT2.18 (incorporated by reference, see § 1238.5) for spectral conditions and ANSI/NAPM IT2.19 (incorporated by reference, see § 1238.5) for geometric conditions for transmission density.

- (i) Recommended visual diffuse transmission background densities for images of documents are as follows:

Expand
Table

Classification	Description of document	Background density
Group 1	High-quality, high contrast printed book, periodicals, and black typing	1.3-1.5
Group 2	Fine-line originals, black opaque pencil writing, and documents with small high contrast printing	1.15-1.4
Group 3	Pencil and ink drawings, faded printing, and very small printing, such as footnotes at the bottom of a printed page	1.0-1.2
Group 4	Low-contrast manuscripts and drawing, graph paper with pale, fine-colored lines; letters typed with a worn ribbon; and poorly printed, faint documents	0.8-1.0
Group 5	Poor-contrast documents (special exception).	0.7-0.85

- (ii) Recommended visual diffuse transmission densities for computer generated images are as follows:

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Table

Film type	Process	Density measurement method	Min. Dmax 1	Max. Dmin 1	Minimum density difference
Silver gelatin	Conventional	Printing or diffuse	0.75	0.15	0.60

Silver gelatin	Full reversal	Printing	1.50	0.20	1.30
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¹ Character or line density, measured with a microdensitometer or by comparing the microfilm under a microscope with an image of a known density.

- (3) **Base plus fog density of microfilms.** The base plus fog density of unexposed, processed microfilms must not exceed 0.10. When a tinted base film is used, the density will be increased. The difference must be added to the values given in the tables in [paragraph \(d\)\(2\)](#) of this section.
- (4) **Line or stroke width.** Due to optical limitations in most micrographic systems, microfilm images of thin lines appearing in the source documents will tend to fill in as a function of their width and density. Therefore, as the reduction ratio of a given system is increased, reduce the background density as needed to ensure that the copies will be legible.

◉ § 1238.16 What are the microfilming requirements for temporary records, duplicates, and user copies?

- (a) **Temporary records with a retention period over 99 years.** Agencies must use the microfilming requirements in [§ 1238.14](#).
- (b) **Temporary records to be kept for less than 99 years, duplicates, and user copies.** NARA does not require the use of specific standards for these microforms. Agencies may select a film stock that meets their needs and ensures the preservation of the microforms for their full retention period. NARA recommends that agencies consult appropriate standards, available as noted in [§ 1238.3](#), and manufacturer's instructions for processing production, and maintenance of microform to ensure that the images are accessible and usable for the entire retention period of the records.

◉ Subpart C—Storage, Use, and Disposition of Microform Records

◉ § 1238.20 How must microform records be stored?

- (a) **Permanent and unscheduled records.** Agencies must store permanent and unscheduled microform records under the extended term storage conditions specified in ISO 18911 and ANSI/PIMA IT9.2 (both incorporated by reference, see [§ 1238.5](#)), except that the relative humidity of the storage area must be a constant 35 percent RH, plus or minus 5 percent. Non-silver copies of microforms must be maintained in a different storage area than are silver gelatin originals or duplicate copies).
- (b) **Temporary records.** Agencies must store temporary microform records under conditions that will ensure their preservation for their authorized retention period. NARA suggests that agencies may consult Life Expectance (LE) guidelines in ISO 18901 (incorporated by reference, see [§ 1238.5](#)).

◉ § 1238.22 What are the inspection requirements for permanent and unscheduled microform records?

- (a) Agencies must inspect, or arrange for a contractor or NARA to inspect master microform of permanent or unscheduled records following the inspection requirements in [paragraph \(b\)](#) of this section.
- (b) The microforms listed in [paragraph \(a\)](#) of this section must be inspected initially in accordance with ANSI/AIIM MS45 (incorporated

by reference, see § 1238.5). All microforms must be inspected when they are two years old. After the initial two-year inspection, unless there is a catastrophic event, the microforms must be inspected as follows until they are transferred to NARA:

- (1) For microfilm produced after 1990, inspect the microfilm every 5 years.
- (2) For microfilm produced prior to 1990, inspect the microfilm every 2 years.
- (c) To facilitate inspection, the agency must maintain an inventory that lists each microform series or publication by production date, producer, processor, format, and results of previous inspections.
- (d) The inspection must include the following elements:
 - (1) An inspection for aging blemishes following ANSI/AIIM MS45 (incorporated by reference, see § 1238.5);
 - (2) A rereading of resolution targets;
 - (3) A remeasurement of density; and
 - (4) A certification of the environmental conditions under which the microforms are stored, as specified in § 1238.20(a).
- (e) The agency must prepare an inspection report, and send a copy to NARA in accordance with § 1238.28(c). The inspection report must contain:
 - (1) A summary of the inspection findings, including:
 - (i) A list of batches by year that includes the identification numbers of microfilm rolls and microfiche in each batch;
 - (ii) The quantity of microforms inspected;
 - (iii) An assessment of the overall condition of the microforms;
 - (iv) A summary of any defects discovered, e.g., redox blemishes or base deformation; and
 - (v) A summary of corrective actions taken.
 - (2) A detailed inspection log created during the inspection that contains the following information:
 - (i) A complete description of all records inspected (title; roll or fiche number or other unique identifier for each unit of film inspected; security classification, if any; and inclusive dates, names, or other data identifying the records on the unit of film);
 - (ii) The date of inspection;
 - (iii) The elements of inspection (see paragraph (d) of this section);
 - (iv) Any defects uncovered; and
 - (v) The corrective action taken.
- (f) If an inspection finds that a master microform is deteriorating, the agency must make a silver duplicate in accordance with § 1238.14 to replace the deteriorating master. The duplicate microform must meet inspection requirements (see § 1238.22) before it may be transferred to a record center or NARA.
- (g) Inspections must be conducted in environmentally controlled areas in accordance with ANSI/AIIM MS45 (incorporated by reference, see § 1238.5).

NARA recommends, but does not require, that agencies use the inspection procedures described in [§ 1238.22\(a\)](#).

⦿ **§ 1238.26 What are the restrictions on use for permanent and unscheduled microform records?**

- (a) Agencies must not use the silver gelatin master microform or duplicate silver gelatin microform of permanent or unscheduled records created in accordance with [§ 1238.14 of this part](#) for reference purposes. Agencies must ensure that the master microform remains clean and undamaged during the process of making a duplicating master.
- (b) Agencies must use duplicates for:
 - (1) Reference;
 - (2) Further duplication on a recurring basis;
 - (3) Large-scale duplication; and
 - (4) Distribution of records on microform.
- (c) Agencies retaining the original record in accordance with an approved records disposition schedule may apply agency standards for the use of microform records.

⦿ **§ 1238.28 What must agencies do when sending permanent microform records to a records storage facility?**

Agencies must:

- (a) Follow the procedures in [part 1232 of this chapter](#) and the additional requirements in this section.
- (b) Package non-silver copies separately from the silver gelatin original or silver duplicate microform copy and clearly label them as non-silver copies.
- (c) Include the following information on the transmittal (SF 135 for NARA Federal Records Centers), or in an attachment to the transmittal. For records sent to an agency records center or commercial records storage facility, submit this information to NARA as part of the documentation required by [§ 1232.14 of this subchapter](#):
 - (1) Name of the agency and program component;
 - (2) The title of the records and the media and format used;
 - (3) The number or identifier for each unit of microform;
 - (4) The security classification, if any;
 - (5) The inclusive dates, names, or other data identifying the records to be included on a unit of microform;
 - (6) Finding aids that are not contained in the microform; and
 - (7) The inspection log forms and inspection reports required by [§ 1238.22\(e\)](#).
- (d) Agencies may transfer permanent microform records to a records storage facility meeting the storage requirements in [§ 1232.14\(a\)](#) (see [§ 1233.10 of this subchapter](#) for NARA Federal Records Centers) of this subchapter only after the first inspection or with certification that the microforms will be inspected by the agency, a contractor, or a NARA Federal Records Center (on a reimbursable basis) when the microforms become 2 years old.

⦿ **§ 1238.30 What must agencies do when transferring permanent microform records to the National Archives of the United States?**

Agencies must:

- (a) Follow the procedures in [part 1235 of this subchapter](#) and the additional requirements in this section.
- (b) If the records are not in a NARA Federal Records Center, submit the information specified in [§ 1232.14\(c\) of this subchapter](#).
- (c) Transfer the silver gelatin original (or duplicate silver gelatin microform created in accordance with [§ 1238.14](#)) plus one microform copy.
- (d) Ensure that the inspections of the microforms are up-to-date. NARA will not accession permanent microform records until the first inspection has been performed (when the microforms are 2 years old).
- (e) Package non-silver copies separately from the silver gelatin original or silver duplicate microform copy and clearly label them as non-silver copies.

§ 1238.32 Do agencies need to request NARA approval for the disposition of all microform and source records?

- (a) ***Permanent or unscheduled records.*** Agencies must schedule both source documents (originals) and microforms. NARA must approve the schedule, SF 115, Request for Records Disposition Authority, in accordance with [part 1225 of this subchapter](#) before any records, including source documents, may be destroyed.
 - (1) Agencies that comply with the standards in [§ 1238.14](#) must include on the SF 115 the following certification: “This certifies that the records described on this form were (or will be) microfilmed in accordance with the standards set forth in [36 CFR part 1238](#).”
 - (2) Agencies using microfilming methods, materials, and procedures that do not meet the standards in [§ 1238.14\(a\)](#) must include on the SF 115 a description of the system and standards used.
 - (3) When an agency intends to retain the silver original microforms of permanent records and destroy the original records, the agency must certify in writing on the SF 115 that the microform will be stored in compliance with the standards of [§ 1238.20](#) and inspected as required by [§ 1238.22](#).
- (b) ***Temporary records.*** Agencies do not need to obtain additional NARA approval when destroying scheduled temporary records that have been microfilmed. The same approved retention period for temporary records is applied to microform copies of these records. The original records can be destroyed once microfilm is verified, unless legal or other requirements prevent their early destruction.

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